

Identification.

Notes for "Applied Microeconometrics".*

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1 Identification and estimation.

Empirical work consists of answering questions of interest (or estimating ‘objects of interest’) given a particular data set. When doing this the empirical analysis can notionally be broken into two separate steps. The first of these is the *identification step*. This is concerned with whether we can find estimates of our objects of interest if we knew what the *population* distributions of the data were. This step does not involve any statistics. The idea is that if we cannot answer the question of interest with the best possible data situation (knowing population values) then we cannot hope to answer it with imperfect information. The second step is the *estimation* (or *inference*) *step*¹. In this stage we are concerned with finding estimates of the population distributions given the finite sample to hand. This involves the usual econometrics and statistics that you have been learning for a while now.

To give a concrete example, suppose the object of interest is the wage distribution of women aged 40 with a university education who were living in England in 2005. Unfortunately, the only population distribution we can estimate (without auxiliary assumptions) is the wage distributions conditional on being employed (since we do not observe the wage for women who are not employed). Knowledge of the latter does not suffice to identify the unconditional wage distribution. This is an example of a ‘selection’ problem which leads to non-identification. We need some extra information or theory to go from the conditional distribution to the unconditional distribution. For example, we could assume that the distribution of those who are not employed is the same as for those who are employed; this is sometimes known as a ‘missing at random’ assumption. This would usually be considered to be an unacceptably

*These notes were prepared for an advanced econometrics MPhil course at Oxford. Please do not quote them without permission.

¹The terminology here is not universally agreed. You may find others who call this step something else or who break it into two steps: estimation and inference.

strong assumption. Alternatively we could use a sample selection estimator that models the participation decision. This requires having some instrument that enters the participation decision but not the wage equation; for example, the presence of children in the household or the husband's wage (conditional on being married). If you go this route, note that we now seem to be modelling something much more difficult (the labour force participation decision) than the original simple question (what is the distribution of wages?). Other possibilities are also available. There is an extensive literature on how to deal with this type of problem.

As another example: it is commonly observed that many individual behaviours are persistent. For example, smoking, taking exercise, arriving at lectures on time etc.. There are a number of candidate explanations. These include: persistence in the environment (people do much the same thing from year to year because their constraints do not change much); state dependence or habit formation (once you start smoking it is difficult to stop); and heterogeneity (we are persistently different). The identifiability question is: what observations would allow us to distinguish between these candidate explanations? For certain, we will need panel data. But what sort?

In practice the idealised two steps of the first paragraph are usually mixed up and most working applied econometricians would not recognise this as a description of what they are doing. For all of us there is a back and forth between the identification step and the estimation step. For example, we might establish identification for a particular set of population distributions and then find that we cannot estimate these consistently with the data we actually have. Then we have to go back to the identification step and find out what new assumptions we have to make to identify our 'objects of interest'. Sometimes this involves revising what we are interested in - in fact, sometimes we end up deciding that we are actually interested in what we can identify!

In these notes we shall only be concerned with the identification step. To consider the parts of this step that are of interest to microeconomicians, it is useful to take a big step back and look at the wood before we examine the trees. The next section may seem excessively abstract but recent experience has suggested that understanding the very broad issues really helps in understanding the narrower ones which directly concern us.

2 Inverse problems.

The *forward problem* (or *direct problem*) is: given a (theoretical) model, what are the implications for observables? In economics, answering this is usually a job for microeconomic theorists (but good applied micro-econometricians will also have to do it some of the time).

The *inverse problem* is: given observations, what is driving them? A classic example of the latter is Newton's development of the 'model' of gravity. The given observables were the parsimonious descriptions of the motions of the planets from Kepler. Newton was better than most at inverse problems like this. In

fact, Newton's research is quite instructive. He faced two inverse problems. The first is whether there was any model that was consistent with all of his data. Having established that there was and that this involved a single unknown parameter, g , he then had a more usual (parametric) identification problem: is the value of g identified from the data? As it happens, we cannot identify g from observations of the movements of the planets; identification required earthly experimental data on falling objects.

Other examples of inverse problems are division (the forward operation is multiplication) and integration (the forward operation is differentiation).²

The standard series of questions concerning inverse problems are:

1. is there any answer?
2. if there is an answer, is it unique?
3. if there is a unique inverse, is it continuous?³

A problem which has answer 'yes' for all three is called a *well-posed inverse problem*. If the answer is no to one of the questions, then we have an *ill-posed inverse problem*.

For the differentiation and integration example, suppose we ask: if the derivative of $f(x)$ is $1/x$ for $x > 0$, what is $f(x)$? The answer to the first question is positive: there is a function which has this derivative, for example $\ln(x)$. The answer to the second question is negative: $4 + \ln(x)$ is also an inverse solution. Thus indefinite integration is an ill posed inverse problem. If we put a side condition on the problem ('make stronger assumptions'), we may end up with a well posed inverse problem. Suppose, for example that the question is: if the derivative of $f(x)$ is $1/x$ for $x > 0$ and $f(1) = 3$, what is $f(x)$? Then $f(x) = 3 + \ln(x)$ is a unique, continuous answer.

Here is an example of an inverse problem in microeconomics. The direct problem is: given a utility function and a set of prices and incomes, what are the demands? The forward problem is conceptually simple - we set the marginal rates of substitution equal to relative prices and find the solution to those $n - 1$ equations and the budget constraint. It is relatively simple to show that these demands will satisfy the four Slutsky conditions⁴ (that is, the latter are necessary conditions on demands). The inverse problem asks:

- given a set of demand functions (quantities as a function of prices and total expenditure), is there a utility function that 'generates' these demands?
- The answer is yes, if the observed demands satisfy the Slutsky conditions.

²It will be clear from this that what is 'forward' and what is 'inverse' is somewhat arbitrary. This small selection suggests that we usually denote the difficult direction as being the inverse problem.

³This is considered important since we do want small changes in the data to lead to large changes in the estimates. As well, continuity usually allows us to easily find an estimator from answering question 2.

⁴Adding-up, zero homogeneity and symmetry and negative semi-definiteness of the Slutsky matrix.

Thus the Slutsky conditions are also sufficient for ‘rationality’. It is quite difficult to show this.

- given that the demands satisfy the Slutsky conditions, is there a unique utility function that is consistent with these demands? The answer is yes and no. The utility function itself is not unique but the preference ordering is unique. That is, the demand functions identify a unique utility function, up to a strictly increasing transformation.
- if we change the observed demands a little, do we still have much the same utility function? Again the answer is yes (if preferences are concave). If we change the demand functions by a small amount (while maintaining the Slutsky conditions) the preferences that are identified will only change by a ‘small amount’.

Note that the inverse problem in this context is known as the *integrability problem* since we differentiate to go from the utility function to demands so when we go backwards we integrate.

The most formal inverse problems you will have seen are probably to do with linear equations. For example, suppose you are given an $n \times m$ matrix A and the equations:

$$Ax = b \tag{1}$$

The forward problem is: ‘given A and x , find b ’. This is simple enough. The inverse problem is: ‘given A and b , find x ’. Generically⁵, there are three cases:

1. If we have more equations than unknowns ($n > m$) then there is (usually) no inverse solution.⁶ That is, the first inverse problem question (usually) has a negative answer.
2. If we have more unknowns than equations ($m > n$) we (usually) have an infinite number of solutions. Thus question 1 is answered yes but question 2 is no.
3. If we have as many equations as unknowns ($n = m$) then we have a unique solution, so long as A is non-singular. In this case the answer to all three questions is ‘yes’ and we have a well posed inverse problem.

When we move to infinite dimensional linear equations things become much less clear. See, for example,

$$\textit{Functional Analysis by Lebedev, Vorovich and Gladwell (2002)} \tag{2}$$

⁵This means that what follows is usually the case, but there are exceptions that need careful noting.

⁶The ‘usually’ here is because the equations do not *necessarily* have to be inconsistent. For example the system of equations with $2x = 4$ and $3x = 6$ has $n = 2$ and $m = 1$ and yet there is a solution ($x = 2$).

Unfortunately, such problems arise quite naturally in many microeconomic applications. For example, we sometimes encounter integral equations such as:

$$g(y) = \int K(y, x) f(x) dx \quad (3)$$

where $g(y)$ and $K(y, x)$ are known and we wish to ‘solve for’ $f(x)$. A solution does not always exist and if it does it may not be unique or continuous. What to do if we have an ill posed inverse problem depends very much on the context. There are no globally useful solutions. That there is something to think about is suggested by the fact that there are *two* mathematics journals devoted solely to the analysis of ill-posed inverse problems!

3 Identification in linear models.

3.1 An ill posed inverse problem.

Suppose we have four random variables (y, x_1, x_2, e) which all have zero mean. Somehow we know that the relationship between the four variables is given by the linear equation:

$$y = \beta x_1 + \gamma x_2 + e \quad (4)$$

We wish to estimate the *parameters of interest*: (β, γ) - this first requires an identification step which is concerned with population values. We assume that we ‘observe’ the following six (population) variances and covariances:

$$E(y^2), E((x_1)^2), E((x_2)^2), E(yx_1), E(yx_2), E(x_1x_2) \quad (5)$$

The important point about what we observe is that we do not assume that we have the population values for anything to do with e . These six population values can be consistently estimated by *analogue estimators*. For example, if we are given a ‘random’ sample of size n on (Y, X_1, X_2) then the fourth value, for example, is consistently estimated by:

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})(X_{1i} - \bar{X}_1) \rightarrow E(yx_1) \text{ as } n \rightarrow \infty \quad (6)$$

If the sample is ‘non-random’ then some other estimator needs to be found or the population values that can be estimated consistently need to be used in this analysis. For example, suppose we only observe a unit if $y > 0$; in this case we can only estimate $E(yx_1 | y > 0)$ which is not usually equal to $E(yx_1)$. Finding estimates of the population values is the *estimation step* - we shall not discuss it any more.

The *identification step* should only concern itself with population values. To see the identification issue, multiply both sides of 4 by x_1 and take expectations (and similarly for x_2):

$$E(yx_1) = \beta E((x_1)^2) + \gamma E(x_1x_2) + E(ex_1) \quad (7)$$

$$E(yx_2) = \beta E(x_1x_2) + \gamma E((x_2)^2) + E(ex_2) \quad (8)$$

We now have four unknowns: $(\beta, \gamma, E(ex_1), E(ex_2))$ and only two equations. Clearly we cannot solve for the parameters of interest uniquely. For example, we could set $\beta = \gamma = 0$ and $y = e$. This is an ill posed problem since we have a solution but it is not unique.

We need further assumptions (apart from the linearity given in 4). We could have taken other variables to multiply both sides of the equation (for example, $(x_1)^2$) but this simply adds one more unobserved term involving e for each new equation. (You should try this). Thus identification requires some assumptions. We now consider four cases.

3.2 Ordinary Least Squares.

The most widely used assumption is:

$$E(ex_1) = E(ex_2) = 0 \quad (9)$$

That is, the observable right hand side variables are uncorrelated with the unobservable right hand side variable (the ‘error term’ e). A shorthand way of saying this is that x_1 and x_2 are *exogenous*. Note this is a theory assumption; somewhere you must have some rationale for saying that the unobservable error term is uncorrelated with each of the observable rhs variables. If we impose the restrictions from (9) then we can solve for the parameters of interest using 7 and 8 (since we now have only two unknowns) if and only if the following matrix is non-singular:

$$\Xi = \begin{bmatrix} E((x_1)^2) & E(x_1x_2) \\ E(x_1x_2) & E((x_2)^2) \end{bmatrix} \quad (10)$$

That is,

$$E((x_1)^2) E((x_2)^2) - (E(x_1x_2))^2 \neq 0 \quad (11)$$

This is the usual ‘full rank’ condition for OLS. Note that this condition fails if and only if x_2 is a linear function of x_1 ($x_2 = kx_1$). If the left hand side of 11 is close to but not equal to zero then the parameters of interest are formally identified, but we may have ‘unstable’ estimates (since we divide by the left hand side of 11 when we solve the equations). If condition 11 holds then the solution of the equations gives the usual OLS estimator:

$$\begin{bmatrix} \hat{\beta} \\ \hat{\gamma} \end{bmatrix} = \Xi^{-1} \begin{bmatrix} E(yx_1) \\ E(yx_2) \end{bmatrix} \quad (12)$$

This two values are unique and continuous in the population values given in (5).

One important thing to note here is that condition (11) is in terms of population values. You may have seen the rank condition written as ‘the data matrix

$$\begin{bmatrix} \sum_{i=1}^n (x_{1i})^2 & \sum_{i=1}^n (x_{1i}x_{2i}) \\ \sum_{i=1}^n (x_{1i}x_{2i}) & \sum_{i=1}^n (x_{2i})^2 \end{bmatrix} = X'X \quad (13)$$

has full rank'. This is mixing up the identification step and the estimation step. For example, Ξ could be singular and yet, for a finite sample, $X'X$ is non-singular. To see this note that if we take *any* two n vectors of the same size, they will generally be population uncorrelated but the sample correlation is very unlikely to be exactly zero.

3.3 One endogenous variable, one instrument.

Suppose that we cannot accept the OLS assumptions. For example, suppose we have reason to believe that x_1 is *endogenous* and x_2 is exogenous:

$$E(ex_1) \neq 0, E(ex_2) = 0 \quad (14)$$

For example, if y is log wages and x_1 and x_2 are schooling and tenure respectively, then no one accepts $E(ex_1) = 0$ since e partly captures unobservable 'ability' which affects the level of schooling directly.⁷ In equations 7 and 8 we now have three unknowns and once again the parameters of interest are not identified. Suppose now that we have another variable z that is assumed to be uncorrelated with the error term:

$$E(ez) = 0 \quad (15)$$

This is known as an *instrumental variable*.⁸ The condition (15) is termed an *exclusion restriction* since it formally implies that variable z does not have a direct effect on y , once we allow for the effects of x_1 and x_2 ; that is, it can be excluded from the equation of interest. Sometimes there is a formal theory to justify the exclusion restriction and sometimes some informal justification - 'I can't think of any reason why z would affect y directly, given that we have conditioned on the other variables in the equation'.

However we find a variable that can be excluded, we assume that we know (can consistently estimate) the covariances for z :

$$E(z^2), E(yz), E(zx_1), E(zx_2) \quad (16)$$

We can now multiply both sides of 4 by z and x_2 and take expectations:

$$\begin{aligned} E(yz) &= \beta E(zx_1) + \gamma E(zx_2) + E(ez) \\ &= \beta E(zx_1) + \gamma E(zx_2) \end{aligned} \quad (17)$$

$$\begin{aligned} E(yx_2) &= \beta E(x_1x_2) + \gamma E((x_2)^2) + E(ex_2) \\ &= \beta E(x_1x_2) + \gamma E((x_2)^2) \end{aligned} \quad (18)$$

⁷In the discussion here I am always taking the coefficients β and γ to be the same for everyone. A recent and lively literature emphasises that things are quite different if we have heterogeneous parameters.

⁸Strictly we should say, a *potential* instrumental variable. The zero covariance with the unobservable is part of being a 'good' instrument. The other part will emerge shortly.

Thus we once again have two equations in two unknowns. The parameters of interest, β and γ , are identified if and only if

$$E(zx_1)E\left((x_2)^2\right) \neq (E(zx_2))E(x_1x_2) \quad (19)$$

If we denote the correlation between z and x_i by ρ_{zi} and the correlation between x_1 and x_2 by ρ_{12} then this can be written:

$$\rho_{z1} \neq \rho_{z2}\rho_{12} \quad (20)$$

This assumption has to be justified. Far and away the most common assumption to make is that z is ‘strongly’ correlated with x_1 (so that the left hand side is large (in absolute value)) but only weakly correlated with x_2 (so that ρ_{z2} is small). For example, if the instrument is a dummy for being a ‘treated unit’ in an experiment (as opposed to being a ‘control unit’) that only impacts directly on x_1 then the left hand side is non-zero and the right hand side will be zero, since $\rho_{z2} = 0$ by the experimental design. If (20) holds then the parameters of interest are said to be *just-identified* since we have just as many equations as unknowns.

One important failure of condition (19) is if z is uncorrelated with either of the two right hand side variables, x_1 and x_2 , so that $\rho_{z1} = \rho_{z2} = 0$. This is quite reasonable, otherwise we could generate random numbers and use these as an instrument. Even if (20) holds, if the two sides are close to equal then we have what is called a *weak instrument*. In this case, the identification is fragile since in practice we must estimate the population values from a finite sample. In practice, in the estimation step we can test for condition (20) - this is called a weak instrument test. Thus we see that a ‘good’ instrument should satisfy the orthogonality condition (15) and (20). The former condition is *not* testable; it is a maintained assumption. To reiterate: if either the instrument, z , or the exogenous variable, x_2 , is uncorrelated with the endogenous variable (so that the right hand side of the inequality 20 is zero) then a good instrument is one that is uncorrelated with the error term but correlated with the endogenous variable. In practice, we often (and loosely) talk of a good instrument as being one that is ‘strongly’ correlated with the endogenous variable; this is only strictly correct if we have one right hand side variable (x_2 is not in the equation).

Although it is not strictly anything to do with identification, it is worth pointing out that we can test for the endogeneity of x_1 in this framework. Under the null hypothesis that $E(ex_1) = 0$ (x_1 is exogenous) OLS should give the same parameter estimates as IVE. Of course, using a finite sample we would always find that the OLS and the IV estimator differed. The inference question is whether the difference is ‘significant’. The test for this is known as a Durbin-Wu-Hausman test. If we conclude that x_1 is exogenous then OLS gives more precise estimates than IV and should be used.

3.4 One endogenous variable, two instruments.

We continue to assume that x_1 is endogenous and x_2 is exogenous. Suppose now that we have two instrumental variables z_1 and z_2 that have:

$$E(ez_1) = E(ez_2) = 0 \quad (21)$$

If we do the same as before, but now multiplying 4 by x_2 , z_1 and z_2 we have three equations in two unknowns:

$$\begin{aligned} E(yz_1) &= \beta E(z_1x_1) + \gamma E(z_1x_2) \\ E(yz_2) &= \beta E(z_2x_1) + \gamma E(z_2x_2) \\ E(yx_2) &= \beta E(x_1x_2) + \gamma E((x_2)^2) \end{aligned} \quad (22)$$

Any of the three pairs of these equations gives an estimate of (β, γ) . If we take a finite sample and estimate these covariances, then the three solutions will differ. The parameters of interest are said to be *over-identified*. Instrumental variable estimation is one way of resolving this conflict. General Methods of Moments (GMM) offers a more general resolution. For both these estimators we can form a test of whether the restrictions to effectively make the three equations into two is valid; this is known as an *over-identifying restriction* test. The basic idea of the test is that the three solutions should not be ‘too different’. This requires that there not be a ‘significant’ difference between, for example, the estimate of $E(yx_2)$ and the estimate of:

$$\begin{bmatrix} E(x_1x_2) & E((x_2)^2) \end{bmatrix} \begin{bmatrix} E(z_1x_1) & E(z_1x_2) \\ E(z_2x_1) & E(z_2x_2) \end{bmatrix} \begin{bmatrix} E(yz_1) \\ E(yz_2) \end{bmatrix} \quad (23)$$

This is an estimation step issue. That is, we ask whether we can change the estimates of the population values so the three equations are consistent. Such tests are very valuable for checking the specification. For example, the test would reject the restrictions if one of the instruments or x_2 was in fact correlated with the error term (x_2 is endogenous). If we do not reject the over-identifying restriction test then we can impose the restrictions needed to reduce the three equations to two.

3.5 Two endogenous variables and one instrument.

The final case we consider is when both x_1 and x_2 are correlated with the error term:

$$E(ex_1) \neq 0, E(ex_2) \neq 0 \quad (24)$$

and we have only one instrument z . In this case we have only one equation (4 multiplied by z) and we cannot solve for unique values of (β, γ) . In terms of an inverse problem, the question (‘uniqueness’) fails. In this case we say that the system is *under-identified*. Either you have to find another instrument or impose more theory restrictions (effectively, enough to have some exogeneity).

4 Identification in nonlinear models.

Sometimes we do not want to assume linearity. For example, suppose we hypothesise that y depends on x in the following way:

$$y = g(x) + u \quad (25)$$

where $g(\cdot)$ is the unknown function that we wish to estimate. The analogue of the OLS assumption (9) is the conditional mean independence assumption $E(u | x) = 0$. If we assume this then we can estimate by taking expectations of both sides, conditional on x :

$$E(y | x) = g(x) \quad (26)$$

That is, the nonparametric (population) regression of y on x identifies $g(\cdot)$. As an aside, attention recently has focussed on so-called non-separable problems in which:

$$y = g(x, u) \quad (27)$$

where $g(x, \cdot)$ is strictly increasing in u . To identify $g(\cdot, \cdot)$ we need stronger conditions than (26). If y a continuous variable then we need that x and u be stochastically independent. If y is discrete (for example, a binary choice variable or a count variable) then identification fails.

Suppose now that $E(u | x) \neq 0$ but we have another variable z which has $E(u | z) = 0$. Thus z is a potential instrument since it is uncorrelated with the unobservable. How could we proceed for the additive error model (25)? Just as before, we can take expectations but this time conditional on z :

$$E(y | z) = E(g(x) | z) = \int g(x) f(x | z) dx \quad (28)$$

The left hand side is a ‘reduced form’ equation except that it differs from the linear case in being the nonparametric regression of y on z . This is potentially observable. We can also estimate the conditional distribution $f(x | z)$. Given that we can estimate these two population objects (one a conditional mean and the other a conditional density), the question is: can we find a unique $g(\cdot)$ function that satisfies this integral equation? The answer is, maybe. We need to put conditions on $f(x | z)$ which are analogous to the ‘strong instrument’ assumptions in the linear model.

The presence of integrals in the nonlinear IV example makes it look quite complicated. A simpler version can be used if we assume that x and z only take on discrete values. That is, $x = x_i$ for $i = 1, \dots, I$ and $z = z_j$ for $j = 1, \dots, J$. In this case, (28) can be written:

$$E(y | z_j) = \sum_{i=1}^I g(x_i) pr(x_i | z_j) \quad (29)$$

where $pr(x_i | z_j)$ is the conditional probability that $x = x_i$ if $z = z_j$. If we denote $E(y | z_j) = y_j$, $g(x_i) = g_i$ and $pr(x_i | z_j) = p_{ji}$, then the y_j 's and p_{ji} 's

are known and the g_i 's are unknown. We have J equations in I unknowns:

$$\begin{bmatrix} y_1 \\ \cdot \\ \cdot \\ y_J \end{bmatrix} = \begin{bmatrix} p_{11} & \cdot & \cdot & p_{1I} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ p_{J1} & \cdot & \cdot & p_{JI} \end{bmatrix} \begin{bmatrix} g_1 \\ \cdot \\ \cdot \\ g_I \end{bmatrix} \quad (30)$$

or:

$$\mathbf{y} = P\mathbf{g} \quad (31)$$

This can be solved uniquely if, for example, $J = I$ (the system is just identified) and P is non-singular. Alternatively, if we have more values for the instrument, z , than for x then we have over-identifying restrictions on the value of p_{ij} which need to be tested and imposed in estimation. If $J < I$ then we cannot identify the values $(g_1, g_2, \dots, g_I)$ without further assumptions.

As another example, suppose we are interested in the distribution of a particular variable, x , (for example, income) but we measure it with additive error so that the observed value y is given by:

$$y = x + u \quad (32)$$

where u is unobserved measurement error. We observe the distribution of the observed value, $F_Y(y)$. The object of interest is the distribution of x , $F_X(x)$. It is clear that if we do not put any distributional assumptions on the error then nothing is identified. For example, if $u = y$ then x is zero for all members of the population: $F_X(x) = 0$ for $x < 0$ and $F_X(x) = 1$ for $x \geq 0$. Conversely, we could take $u = 0$ so that the distribution of x is the same as the observed distribution of y : $F_X(\cdot) = F_Y(\cdot)$. If we know the distribution of u then the distribution of x can be found by deconvolution. This is not very helpful since we do not usually know anything about the unobservable. For example, assuming u is Normally distributed with zero mean and an unknown variance imposes an arbitrary functional form. This is known as *identifying off of functional form*. Generally we are looking for *nonparametric identification* - identification that does not depend on arbitrary assumptions regarding functional forms and distributions. Now suppose we have two noisy measures of x , denoted y_1 and y_2 :

$$\begin{aligned} y_1 &= x + u_1 \\ y_2 &= x + u_2 \end{aligned} \quad (33)$$

Then a very useful result by Kotlarski states that if x , u_1 and u_2 are mutually independent then we can identify the distribution of x from observing the joint distribution of (y_1, y_2) . Stochastic independence is a strong assumption but if u_1 and u_2 are really felt to be 'different sorts' of measurement error, neither of which depends on the true value (x) , then it may be justified.

5 Set identification.

All of the above has been about finding *point estimates*. For example, in (4) we wanted estimates of the two parameters β and γ . Sometimes we can settle for having a range of values for the objects of interest. This leads to the idea of *partial identification* or *set identification*. This a very exciting and topical research topic. The book for this is:

Partial identification of probability distributions

by Charles Manski (2003)

In terms of the three inverse problem questions, we are here considering inverse problems which have many solutions.

To give an example from chapter 1, complement 1a from the Manski book, suppose we wish to find the proportion of a given population who are employed at a particular date. Manski considers data from the National Longitudinal Survey of Youth (NLSY). The population consists of people who were born between 1957 and 1964 and who resided in the US in 1979 (the year the panel began). The question of interest is: how many of this population were employed in 1991 (one of the panel survey years). The survey agency originally (in 1979) contacted 6,812 individuals. The assignment to the survey was ‘random’. Of those contacted, 6,111 agreed to be interviewed in 1979. By 1991 some people had dropped out of the panel and information on employment status could be gathered for 5,556 of the sample. Table 1 gives the numbers in the survey. We shall sidestep the estimation step and take these sample proportions to be the population values.

Employment status	Number	Proportion (%)
Employed	4,332	63.6
Unemployed	297	4.36
Out of labour force	927	13.6
Total in panel in 1991	5,556	81.6
Dropped out of NLSY	555	8.15
Never interviewed	701	10.3
Total	6,812	100

Table 1: 1991 employment status, NLSY

If we were looking for point identification we would have to make strong assumptions to identify the proportion employed. For example, we could assume that selection into the group who were not interviewed in 1991 is ‘missing at random’. Then the employment rate is given by $100 * 4332/5556 = 78.0\%$. The problem with this assumption is that it is not credible. For example, busy people who have a higher employment rate might have dropped out of the panel. This would bias the employment rate downwards. Alternatively, the homeless (who are less likely to be employed) might drop out of the panel and this would bias

the employment rate upwards. Of course, we could simply hope that the biases cancel each other out, but it is the expression of hopes like this that give applied microeconometrics a bad name. If we want to do a good job of dealing with this issue with point identification framework we need a theory of why people did not join the survey in 1979 and why some of those that did join, later dropped out. That is, we need a theory of attrition or sample selection. This might give some instruments that affect the probability of being in the sample in 1991 but can be excluded from the determination of employment status. Typically, however, we do not have any information on the people who never joined the survey and we have to make stronger assumptions, such as their reason for never joining are the same as the reasons for the dropping out of some of those who joined in 1979.

An alternative approach is to give up on point identification and ask instead whether we can find bounds on the employment rate. The only theory here is that in 1991 all of those who were not surveyed in 1991 were either in employment or they were not. The maximum number of possible employed is those who reported being employed and all of those who made no report. This gives $4332 + 555 + 701 = 5,588$ with an employment rate of $100 * 5588 / 6812 = 82.0\%$. Conversely, if all of those not interviewed were not employed we would have a rate of $100 * 4332 / 6812 = 63.6\%$. Thus we identify the interval $[63.6, 82.0]$ as giving the bounds on employment rates with no assumptions concerning non-response. This may seem quite a wide interval, but that is the price we pay for ‘losing’ about 18% of the original population. If you feel uneasy about the lower bound assumption (for example, some of those who dropped out did so because they died and they are surely not still employed) then you might start to think of assumptions that tighten the bounds a bit. You can do this and still stay within a partial identification framework. For example, if you could find assumptions you thought acceptable to most people that gave bounds of, say, $[76.0, 82.0]$ that might be enough for whatever you are going to use the estimates for. There is no need to make very strong assumptions to achieve point identification if weak assumptions give reasonably tight bounds.

Most people who have a practical interest in using our estimates do not like bounds. They want a number. In this case, one can use the identified set to give some informal idea of the ‘strength’ of any conditions used for point identification. For example, suppose we assumed that the selection was due to people being ‘missing at random’. Then the point estimate is 78.0%. The gain in precision from going from the bounds $[63.6, 82.0]$ which require no assumptions to the point estimate *are entirely due to our ‘missing at random’ assumption* and do not derive from the data. Other assumptions would give other values (all within the nonparametric bounds).